POULTRY RESEARCH, DIAGNOSTICS AND GENETICS CENTER

A practical phased plan for the Nigeria poultry operation

Purpose: reduce disease and mortality, improve egg and meat productivity, understand bird behavior and housing conditions, produce better feed, and build a long-term breeding program.

Guiding principle
Start with reliable farm records, basic sensors, sound veterinary support and feed quality control. Expand into advanced diagnostics and genetics only when the data and business case justify the investment.

Prepared from our discussions on poultry health, feed formulation, environmental monitoring, farm data, and breeding strategy.

1. Vision and core functions

The center should combine four functions that work together rather than operate as separate projects:

1) Disease diagnostics and prevention — investigate deaths and illness, monitor vaccination and biosecurity, test feed and water, and identify recurring health problems.

2) Nutrition and productivity — evaluate feed ingredients and finished rations; improve egg production, shell quality, broiler growth, feed conversion and cost per unit of output.

3) Genetics and breeding — select birds for survival, productivity, feed efficiency, fertility, hatchability and suitability for Nigerian conditions.

4) Farm research and data — combine video, environmental sensors, feed/water consumption, egg records, body weights and health events to learn what drives performance.

> The central question for every intervention should be: did it measurably improve health, yield, welfare or cost?

2. Phased development plan

Phase 1 — Farm monitoring and veterinary partnership (start here)

• Create daily records for feed consumed, water consumed, eggs laid, egg weight, deaths, culls, symptoms, treatments, vaccination and changes in feed or lighting.

• Install a small camera-and-sensor system in one representative house; keep data locally so internet interruptions do not stop recording.

• Keep retained samples of each feed batch and ingredient lot, with date and supplier labels.

• Use a qualified poultry veterinarian and an established diagnostic lab for necropsies, disease testing and vaccination review.

Phase 2 — Nutrition and diagnostic capacity

• Add a poultry nutritionist and trained laboratory staff. Develop only the tests that can be safely and competently performed on site; refer specialized or high-risk pathogen testing to an authorized laboratory.

• Build a repeatable feed-quality process: ingredient identification, moisture and storage checks, formulation review, batch weighing verification and periodic laboratory analysis.

Phase 3 — Breeding and genetics

• Begin with performance recording and planned selection, not an immediate attempt to create a new commercial strain.

• Keep layer and broiler breeding objectives separate. Use controlled mating, pedigree/parent records, hatch records and offspring performance tests.

• Partner with a university or specialist genetics organization for genomic analysis and advanced breeding design.

3. Cameras, sensors and farm data

Recommended low-cost pilot: start with one house and test the setup before installing it everywhere.

Equipment	What it records / why	Starting setup
PoE IP cameras	Bird activity, movement, crowding, feeding/drinking-area behavior and unusual inactivity. Video is evidence for later review; it does not diagnose disease by itself.	2 cameras per house to start; 4MP or 4K, wide angle, infrared night vision, dust/humidity suitable; record to local NVR.
Temperature/humidity loggers or connected sensors	Heat and humidity patterns where birds actually live; helps investigate heat stress and differences across the house.	Place at bird height in different locations. Record every 1–5 minutes where possible; protect from pecking, dust and direct water.
Lux meter / light sensor	Light intensity at bird level and how lighting differs across the house.	Handheld lux meter is the cheapest start; take repeatable manual readings. Add a connected light sensor later for continuous data.
Water meter	Daily water consumption; a sudden change can be an early warning.	Choose a meter matched to pipe size and flow. Record daily totals; automated logging is a later upgrade.
Feed weighing	Feed offered, feed remaining and estimated daily intake.	Start with a reliable platform scale and a consistent recording method; automate later.
Egg and bird records	Egg count, egg weight, shell defects, mortality and body-weight trends.	Eggs and deaths daily; sample body weights weekly or biweekly, using a consistent method.

How the data system works

Simple offline setup: standalone temperature/humidity loggers save readings internally; download them to a laptop periodically. A handheld lux meter requires manual readings entered in a spreadsheet. Cameras record to a local NVR or storage drive.

Connected setup: compatible sensors send readings to a central controller or gateway; a local computer stores time-stamped data and displays trends/alerts. Internet can be optional if the system stores locally and synchronizes when connectivity returns.

Important: make sure all records use synchronized dates/times and identify the house, flock and sensor location. Without this, camera clips cannot be reliably compared with heat, feed, water and egg records.

4. Initial equipment budget

Approximate discussion-stage estimates in USD; these are planning ranges, not verified live quotations. Shipping, import duties, installation and local availability in Nigeria are excluded.

Item	Estimated cost	Notes
2 PoE cameras	$100–160	Camera prices vary by model; a compatible PoE switch/injector, cabling and mounts may be needed.
2 temperature/humidity loggers	$45–90	Standalone logger option; connected sensors may cost more.
Handheld lux meter	$20–40	Manual readings; no central connection required.
Water meter	$30–100	Must match the pipe and expected flow.
Cables, mounts, storage / accessories	$40–80	Actual cost depends on layout and recording capacity.

Item	Estimated cost	Notes
Indicative total	$235–470	A practical target was $300–400 before shipping.

5. Disease prevention and diagnostics

The aim is to detect problems early and determine the cause rather than guess based on symptoms alone.

• Establish written biosecurity: visitor control, dedicated footwear/equipment, rodent and wild-bird control, cleaning/disinfection, safe carcass disposal and separation of age groups where practical.

• Create a vaccination calendar with a poultry veterinarian based on local disease risks, flock age, product instructions and local requirements.

• Keep a mortality log and record the date, house, age, feed batch, symptoms and recent changes. Sudden increases trigger veterinary escalation.

• For unexplained deaths, arrange prompt necropsy and appropriate samples through a veterinarian. Store/transport samples as instructed by the diagnostic lab.

• Test feed ingredients and finished feed when indicated, including mould/mycotoxin risks in stored grains. Review water quality and sanitation.

• Use antibiotics or other medicines only under veterinary guidance, with correct dosing and withdrawal periods. Do not use supplements as a substitute for diagnosis.

Partnership starting point

The National Veterinary Research Institute (NVRI) was discussed as a potential Nigerian partner for diagnostic services, post-mortem examinations and disease surveillance. Confirm current services, sample requirements and fees directly: nvri.gov.ng.

6. Feed formulation and nutrition quality control

The recent flock incident illustrates why a recipe copied from a premix label or another farm should not be assumed to be balanced. A sharp fall in egg production and deaths require veterinary investigation as well as nutrition review.

• Formulate on nutrient targets—not ingredient percentages alone. Check metabolizable energy, crude protein, digestible/available amino acids (especially methionine and lysine), calcium, available phosphorus, sodium/chloride, vitamins and trace minerals.

• Verify premix brand, intended inclusion rate and compatibility. Do not substitute a 1% additive for a premix labelled for 2% without manufacturer confirmation.

• Check limestone calcium content and particle size; check dicalcium phosphate (DCP) specifications for calcium and phosphorus. Calculate contributions from every ingredient before deciding inclusion rates.

• Use salt only at a calculated rate after accounting for sodium/chloride from all ingredients and premix.

• Weigh each batch accurately, document actual weights, label the batch and retain a representative sample.

• Check grain storage, moisture, mould and mycotoxin risk. Consider laboratory analysis of ingredients and finished feed.

Practical feed workflow

1) Identify suppliers and batch lots → 2) check ingredient quality → 3) have a qualified poultry nutritionist formulate → 4) weigh and mix accurately → 5) verify the finished feed periodically → 6) compare feed intake, egg output, shell quality, mortality and cost.

7. Lighting, behavior and welfare

• Measure lux at bird height in several fixed points, not just near the light source. Map bright and dark areas.

• Record light-on/light-off times and total daily light duration. Avoid sudden lighting changes; set the program with a poultry specialist for flock age and production stage.

• Use video to observe flock distribution, feeding/drinking activity, resting, crowding, feather pecking and unusual inactivity. Review short, repeatable time windows rather than trying to manually watch all footage.

• Interpret behavior together with temperature, humidity, water/feed intake, egg production and health records. A camera pattern alone cannot confirm a disease.

• In hot conditions, look for changes in bird distribution, panting and water use; ensure ventilation, shade and water systems are appropriate to the house design.

8. Productivity and breeding program

Start collecting high-quality records from the birds already on the farm before investing in specialized genetic equipment.

Layer selection traits	Broiler selection traits
Eggs per hen; egg weight; shell strength; survival; feed per dozen eggs; fertility; heat tolerance.	Growth rate; feed conversion; sale weight; breast yield; survival; leg health; heat tolerance.

• Keep layer and broiler breeding objectives separate; their optimal traits differ.

• For breeding candidates, record identity/parentage where known, hatch date, health, body weight, fertility, hatchability and offspring performance.

• Use controlled mating and a documented selection plan to avoid accidental inbreeding. Maintain adequate population size and genetic diversity.

• Compare candidate offspring against a commercial benchmark under similar management and feed conditions.

• Seek specialist support before launching a formal breeding line; competitive genetic improvement takes multiple generations.

9. Suggested data dashboard

Daily	Weekly / periodic	Event-based
Egg count and lay rate; deaths/culls; feed and water used; house temperature/humidity; lighting schedule; notable behavior.	Sample egg weights and shell defects; sample bird weights; feed conversion estimates; camera review summaries; sensor checks.	Feed batch change; vaccination/treatment; equipment failure; power outage; disease event; unusually high mortality or drop in intake/eggs.

10. First 30 days: practical action plan

Week 1: appoint a responsible farm-records lead; create daily flock, feed, water, egg and mortality sheets; document feed batch and supplier information; confirm veterinary contact.

Week 2: install cameras and environmental loggers in one house; take manual lux readings; test the recording and backup power; synchronize timestamps.

Week 3: review feed formula and ingredient quality with a poultry nutritionist; define a sampling plan; inspect water lines, lighting and ventilation.

Week 4: summarize trends and set baseline metrics. Only then decide which automated sensors or lab tests would add enough value to justify their cost.

Key caution

This document is a planning framework, not a veterinary diagnosis, validated feed formula, or facility engineering specification. Disease testing, medicines, vaccination, feed formulation and breeding decisions should be reviewed by appropriately qualified professionals and comply with Nigerian requirements.

Reference and purchasing starting points

- National Veterinary Research Institute: https://nvri.gov.ng/
- Reolink camera range: https://reolink.com/
- Elitech data loggers: https://www.elitecheu.com/

Equipment costs in this plan are indicative estimates from our discussion and should be confirmed with vendors before purchase.